



Lista exercício 2

Calcule

(1) $\int_0^1 (x+3) dx$

(2) $\int_{-1}^1 (2x+1) dx$

(3) $\int_0^4 \frac{1}{2} dx$

(4) $\int_{-2}^1 (x^2-1) dx$

(5) $\int_1^3 dx$

(6) $\int_{-1}^2 4 dx$

(7) $\int_1^3 \frac{1}{x^3} dx$

(8) $\int_{-1}^1 5 dx$

(9) $\int_0^2 (x^2+3x-3) dx$

(10) $\int_0^1 \left(5x^3 - \frac{1}{2}\right) dx$

(11) $\int_1^1 (2x+3) dx$

(12) $\int_1^0 (2x+3) dx$

(13) $\int_{-2}^{-1} \left(\frac{1}{x^2} + x\right) dx$

(14) $\int_0^4 \sqrt{x} dx$

(15) $\int_1^4 \frac{1}{\sqrt{x}} dx$

(16) $\int_0^8 \sqrt[3]{x} dx$

(17) $\int_{-1}^0 \left(x^3 - 2x + 3\frac{1}{x^4}\right) dx$

(18) $\int_1^2 \left(x^3 + x + \frac{1}{x^3}\right) dx$

(19) $\int_1^4 (5x + \sqrt{x}) dx$

(20) $\int_1^2 \frac{1+x}{x^3} dx$

(21) $\int_0^1 (x+1)^2 dx$

(22) $\int_1^4 \frac{1+x}{\sqrt{x}} dx$

(23) $\int_0^2 (t^2 + 3t - 1) dt$

(24) $\int_1^2 \frac{1+t^2}{t^4} dt$

$$(25) \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \cos(2x) \, dx$$

$$(26) \int_{-\pi}^0 \sin(3x) \, dx$$

$$(27) \int_{-1}^1 e^{2x} \, dx$$

$$(28) \int_0^1 \frac{1}{1+x^2} \, dx$$

$$(29) \int_0^{\frac{\pi}{4}} \sin x \, dx$$

$$(30) \int_{-1}^0 e^{-2x} \, dx$$

$$(31) \int_0^{\frac{\pi}{3}} (3 + \cos(3x)) \, dx$$

$$(32) \int_0^2 2^x \, dx$$

$$(33) \int_0^{\frac{\pi}{3}} (\sin x + \sin(2x)) \, dx$$

$$(34) \int_0^{\frac{\pi}{4}} \left(\frac{1}{2} + \frac{\cos(2x)}{2} \right) \, dx$$

$$(35) \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$$

$$(36) \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

Sugestão: Verifique que

$$\cos^2 x = \frac{1 + \cos(2x)}{2} \quad e \quad \sin^2 x = \frac{1 - \cos(2x)}{2}$$

Referência:

[1] Guidorizzi, H.L. Um curso de cálculo vol 1